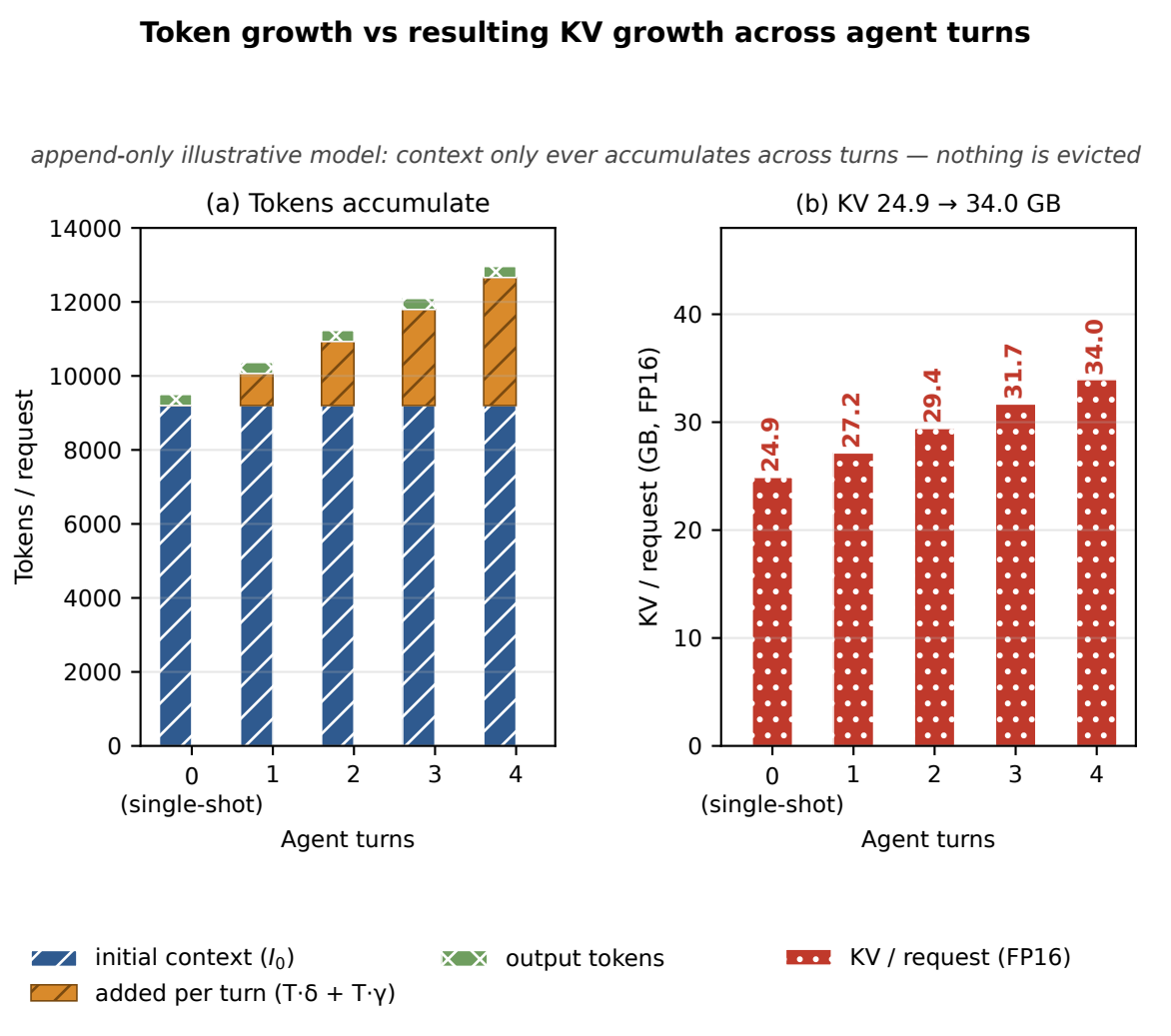
Token growth vs resulting KV growth across agent turns
append-only illustrative model: context only ever accumulates across turns — nothing is evicted
(a) Tokens accumulate
Tokens / request
14000
12000
10000
8000
6000
4000
2000
0
0
(single-shot)
1
2
3
4
Agent turns
(b) KV 24.9 → 34.0 GB
KV / request (GB, FP16)
40
30
20
10
0
24.9
27.2
29.4
31.7
34.0
0
(single-shot)
1
2
3
4
Agent turns
initial context ($I_0$)
added per turn (T·δ + T·γ)
output tokens
KV / request (FP16)